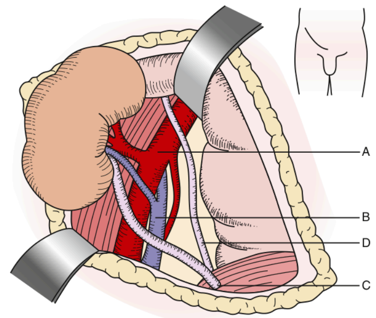

Fig. 69.1 Anatomy of a typical first kidney transplant. (A) External iliac artery, (B) external iliac vein, (C) implanted donor ureter, and (D) native ureter.

Fig_69_1

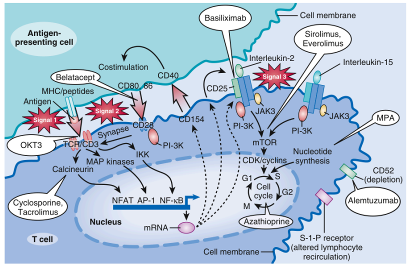

Fig. 69.2 Stages of T-cell activation: multiple targets for immunosuppressive agents. Signal 1: The Ca^{++}-dependent signal induced by T cell receptor (TCR) stimulation results in calcineurin activation, a process inhibited by the calcineurin inhibitors (CNIs). Calcineurin dephosphorylates nuclear factor of activated T cells (NFAT), enabling it to enter the nucleus and bind to the interleukin-2 (IL-2) promoter. Corticosteroids bind to cytoplasmic receptors, enter the nucleus, and inhibit cytokine gene transcription in both the T cell and antigen-presenting cell (APC). Corticosteroids also inhibit activation of the transcription factor, nuclear factor-κB (not shown). Signal 2: Costimulatory signals, such as that between CD28 on the T cell and B7 on the APC, are necessary to optimize T cell transcription of the IL-2 gene, prevent T-cell anergy, and inhibit T cell apoptosis. Signal 3: IL-2 receptor stimulation induces the cell to enter the cell cycle and proliferate. IL-2 and related cytokines have both autocrine and paracrine effects. Signal 3 may be blocked by IL-2 receptor antibodies or by sirolimus. Further downstream, azathioprine and mycophenolate mofetil (MMF) inhibit progression into the cell cycle by inhibiting purine and therefore DNA synthesis. (From Halloran PF. Immunosuppressive drugs for kidney transplantation. *N Engl J Med.* 2004;351:2715–2729.)

Fig_69_2

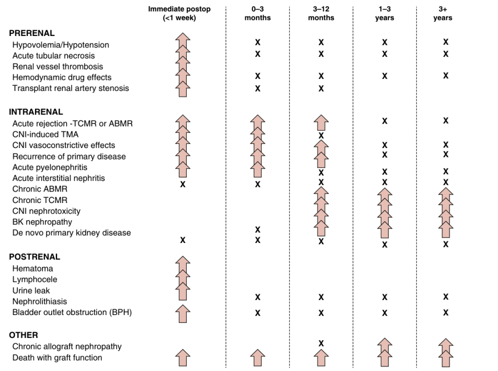

Fig. 69.3 Causes of allograft dysfunction during the posttransplant period. *Arrow,* Increased incidence; *X, occurs,* though less frequently.

Fig_69_3

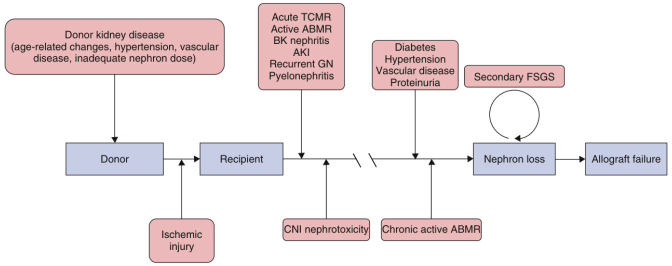

Fig. 69.4 Algorithm for management of persistent delayed graft function. The presence of antidonor human leukocyte antigen (HLA) antibodies should prompt immediate biopsy in this setting.

Fig_69_4

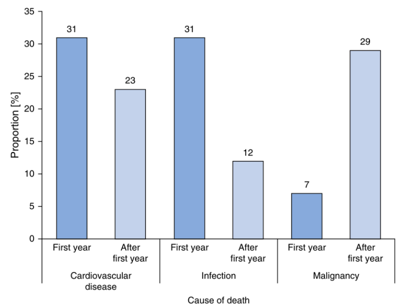

Fig. 69.5 Principal causes of death after kidney transplantation in patients with functioning kidney allografts.

Fig_69_5

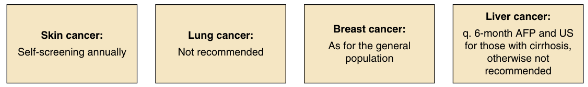

Fig. 69.6 Cancer screening recommendations for kidney transplant recipients. *PAP,* Papanicolaou test; *PSA,* prostate specific antigen; *DRE,* digital rectal exam. *Based on the most common recommendations from the AST (2000), KDIGO (2009), EBPG (2002), CST (2010), NKF (2009) and RA (2011) guidelines.*

Fig_69_6

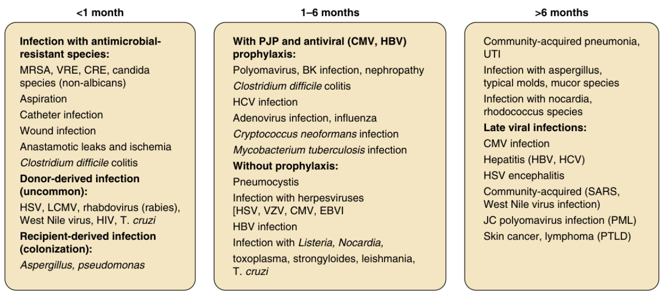

Fig. 69.7 Pattern of infection after transplantation according to time after transplantation.

Fig_69_7

Table 69.1 Surgical Complications of the Transplant Procedure[149,413-415]

Complication	Description	Management
Hemorrhage	Usually involves the anastomotic site. Perirenal hematomas can result from venous or small artery bleeding.	Unless small and stable, surgical exploration is required. Large hematomas should be evacuated in order to reduce the risk for subsequent infection.
Renal artery thrombosis	Rare (<1%) and usually related to an anastomotic problem or kink in the renal artery. Risk factors include recipient arteriosclerosis, multiple donor arteries, vasospasm, and hypotension.	Delays inherently associated with confirming the diagnosis and preparing the patient for surgical exploration usually exceed the time required to reestablish arterial flow to the kidney, resulting in prolonged warm ischemia, hypoxia, and often permanent loss of function.
Renal vein thrombosis	Causes include problems with the surgical anastomosis, extrinsic compression (by a lymphocele/hematoma), or deep venous thrombosis that extends in the iliac vein at the level of the venous anastomosis; thrombophilia may contribute. Allograft ultrasound shows decreased or absent blood flow in the renal vein and either absent/reversed diastolic arterial flow.	Surgical exploration to attempt thrombectomy followed by anticoagulation can be performed but is rarely successful.
Arterial anastomosis pseudoaneurysm	Rare, infectious complication that leads almost invariably to graft loss, associated with significant mortality and morbidity. Though rare, distal thrombosis of the femoral artery or arterial dissection can lead to loss of a lower limb.	Transplant nephrectomy, vascular reconstruction and/or excision with extra anatomic bypass is usually required. If less severe, treatment with covered stenting can be attempted.
Lymphocele	Lymphatic fluid collection originating from either the severed iliac lymphatics or lymphatic drainage of the renal allograft itself. Analysis of aspirated fluid will typically show a high lymphocyte count and a creatinine concentration similar to serum.	Lymphoceles frequently reaccumulate following drainage, although injection of sclerosing agent following aspiration may reduce the risk of recurrence. The preferred and more definitive treatment is internal drainage of the lymphocele into the peritoneal cavity. In many centers, a laparoscopic transabdominal approach has replaced the traditional open approach that utilizes the kidney transplant incision.
Urinoma/urine leak	Causes include infarction of the ureter due to perioperative disruption of its blood supply (typically a lower pole artery infarction) and breakdown of the ureterovesical anastomosis. Analysis of aspirated fluid will typically show a fluid creatinine concentration much higher than serum.	A bladder catheter should be immediately inserted to decompress the urinary tract. Many cases require surgical exploration and repair. The type of repair depends on the level of the leak and viability of involved tissues.

Table_69_1

Table 69.2 Drugs Used in Maintenance Immunosuppression

Drug	Mechanism of Action	Adverse Effects
Corticosteroids	Block synthesis of several cytokines including IL-2; multiple antiinflammatory effects	Glucose intolerance, hypertension, hyperlipidemia, osteoporosis, osteonecrosis, myopathy, cosmetic defects, growth suppression in children
Cyclosporine	Inhibits calcineurin (binds cyclophilin)-dependent synthesis of IL-2 and other molecules critical for T cell activation; T cell activation thereby inhibited	Nephrotoxicity (acute and chronic), hypertension (more than tacrolimus), posttransplant diabetes, neurotoxicity, hyperkalemia, hypomagnesemia, metabolic acidosis, hypercalciuria, and dyslipidemia (more than tacrolimus). Hypertrichosis, gingival hyperplasia (particularly when used in combination with dihydropyridine calcium channel blockers)
Tacrolimus	Similar to cyclosporine, though binds FKBP	Similar to cyclosporine, though more hyperkalemia, neurotoxicity, posttransplant diabetes, and alopecia
Azathioprine	Inhibits purine biosynthesis; lymphocyte replication therefore inhibited	Bone marrow suppression, rarely pancreatitis, hepatitis
Mycophenolate mofetil	Inhibits de novo pathway of purine biosynthesis (relatively lymphocyte selective); lymphocyte replication therefore inhibited	Bone marrow suppression, gastrointestinal upset, invasive CMV disease more common than with azathioprine
Sirolimus	Sirolimus-FKBP complex inhibits TOR blocking lymphocyte proliferative response	Bone marrow suppression, proteinuria, mouth ulcers, hyperlipidemia, interstitial pneumonitis, edema, enhanced nephrotoxicity of cyclosporine/tacrolimus
Belatacept	Blocks T cell costimulation	PTLD in EBV seronegative, PML (rare), reactivation of TB

CMV, Cytomegalovirus; *EBV,* Epstein-Barr virus; *FKBP,* FK-binding protein; *IL-2,* interleukin-2; *PTLD,* posttransplant lymphoproliferative disorder; *TB,* tuberculosis; *TOR,* target of rapamycin.

Table_69_2

Table 69.3 Agents That May Interact With Transplant Medications

Drugs That Interact With Calcineurin Inhibitors and Sirolimus

Class of Drug	Increase Level	Decrease Level
Ca channel blocker	Diltiazem, verapamil	
Antibiotics	Erythromycin, azithromycin, clarithromycin	Nafcillin
Antifungals	Fluconazole, ketoconazole, itraconazole, voriconazole	
Antituberculin		INH, rifampin, rifabutin
Antiviral	Ritonavir, nelfinavir, saquinavir	Efivrez, nevirapine
Antiseizure		Phenytoin, phenobarbital, carbamazepine, primidone
Antidepressant	Fluoxitine, nefazodone, fluvoxamine	
Foods and Herbal Preparations That Interact With Calcineurin Inhibitors		
Food	Grapefruit juice/pomegranate juice	
Herbs		St. Johns wort

Table_69_3

Table 69.4 Updates of 2019 Banff Classification for Antibody-Mediated Rejection; Borderline Changes, T Cell–Mediated Rejection (TCMR), and Polyomavirus Nephropathy

Category 1: Normal biopsy or nonspecific changes
Category 2: Antibody-mediated changes
Active ABMR: all 3 criteria must be met for diagnosis
1. Histologic evidence of acute tissue injury, including 1 or more of the following:
- Microvascular inflammation (g >0 and/or ptc >0), in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥1 alone is not sufficient and g must be ≥1
- Intimal or transmural arteritis (v >0)b
- Acute thrombotic microangiopathy, in the absence of any other cause
- Acute tubular injury, in the absence of any other apparent cause

2. Evidence of current/recent antibody interaction with vascular endothelium, including 1 or more of the following:
- Linear C4d staining in peritubular capillaries or medullary vasa recta (C4d2 or C4d3 by IF on frozen sections, or C4d >0 by IHC on paraffin sections)
- At least moderate microvascular inflammation ([g + ptc] ≥2) in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥2 alone is not sufficient and g must be ≥1
- Increased expression of gene transcripts/classifiers in the biopsy tissue strongly associated with ABMR, if thoroughly validated

3. Serologic evidence of circulating donor-specific antibodies (DSA to HLA or other antigens). C4d staining or expression of validated transcripts/classifiers as noted above in criterion 2 may substitute for DSA; however, thorough DSA testing, including testing for non-HLA antibodies if HLA antibody testing is negative, is strongly advised whenever criteria 1 and 2 are met

Chronic active ABMR; all 3 criteria must be met for diagnosis
1. Morphologic evidence of chronic tissue injury, including 1 or more of the following:

Transplant glomerulopathy (eg >0) if no evidence of chronic TMA or chronic recurrent/de novo glomerulonephritis; includes changes evident by electron microscopy (EM) alone (cg1a)
Severe peritubular capillary basement membrane multilayering (ptcml1; requires EM)
Arterial intimal fibrosis of new onset, excluding other causes; leukocytes within the sclerotic intima favor chronic ABMR if there is no prior history of TCMR, but are not required
2. Identical to criterion 2 for active ABMR, above
3. Identical to criterion 3 for active ABMR, above, including strong recommendation for DSA testing whenever criteria 1 and 2 are met. **Biopsies meeting criterion 1 but not criterion 2 with current or prior evidence of DSA (posttransplant) may be stated as showing chronic ABMR; however, remote DSA should not be considered for diagnosis of chronic active or active ABMR**

Chronic (inactive) ABMR
1. cg >0 and/or severe ptcml (ptcml1)

C4d staining without evidence of rejection; all 4 features must be present for diagnosis
1. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d >0 by IHC on paraffin sections)
2. Criterion 1 for active or chronic active ABMR not met
3. No molecular evidence for ABMR as in criterion 2 for active and chronic active ABMR
4. No acute or chronic active TCMR. or borderline changes

Category 3: Borderline (suspicious) for acute TCMR
Foci of tubulitis (tl. t2. or t3) with **mild interstitial inflammation (i1)**, or mild (t1) tubulitis with moderate-severe interstitial inflammation (i2 or i3)
No intimal or transmural arteritis (v = 0)
Category 4: TCMR
Acute TCMR
Grade IA: Interstitial inflammation involving >25% of nonsclerotic cortical parenchyma (i2 or i3) with moderate tubulitis (t2) involving 1 or more tubules, not including tubules that are severely atrophied
Grade IB: Interstitial inflammation involving >25% of non-sclerotic cortical parenchyma (i2 or i3) with severe tubulitis (t3) involving 1 or more tubules, not including tubules that are severely atrophic
Grade IIA: Mild to moderate intimal arteritis (v1), with or without interstitial inflammation and/or tubulitis
Grade IIB: Severe intimal arteritis (v2), with or without interstitial inflammation and/or tubulitis
Grade III: Transmural arteritis and/or arterial fibrinoid necrosis involving medial smooth muscle with accompanying mononudear cell intimal arteritis (v3), with or without interstitial inflammation and/or tubulitis
Chronic active TCMRe
Grade IA: Interstitial inflammation involving >25% of sclerotic cortical parenchyma (i-IFTA2 or i-IFTA3) AND >25% of total cortical parenchyma (ti2 or ti3) with moderate tubulitis (t2 or **t-IFTA2**) involving 1 or more tubules, not including severely atrophic tubules: other known causes of i-IFTA should be ruled out
Grade IB: Interstitial inflammation involving >25% of sclerotic cortical parenchyma (i-IFTA2 or i-IFTA3) AND >25% of total cortical parenchyma (ti2 or ti3) with severe tubulitis (t3 or **t-IFTA3**) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
Grade II: Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima). This may also be a manifestation of chronic active or chronic ABMR or mixed ABMR/TCMR
Category 5: polyomavirus nephropathyf
PVN Class 1
pvl 1 and ci 0-1
PVN Class 2

[a]All updates in boldface type.
From Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney Meeting Report (I): Updates on and clarification of criteria for T cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20(9):2318–2331.

Table_69_4

Table 69.5 Recurrent Glomerulonephritis After Kidney Transplantation[274,277,416,417]

Primary Disease	Recurrence Rate	Graft Loss With Recurrence	Presentation	Clinical Predictors of Recurrence	Treatment*
Primary FSGS	20-50% for primary; rare for familial	13-20% at 10 years	Can manifest as massive proteinuria (nephrotic range in 80%) within hours to weeks post transplant, hypertension, graft dysfunction	White ethnicity Younger age Rapidly progressive FSGS in primary course Recurrence (especially early) in a prior transplant Living related donors (controversial)	Preemptive perioperative plasmapheresis for 1-2 weeks pretransplant for patients at high risk Early plasmapheresis for relapse, poor prognosis if delayed Consider high-dose CNIs, ACE inhibitors, high-dose steroids, cyclophosphamide, rituximab
Anti-GBM Disease	Rare if anti-GBM antibody negative x 6 months at transplant	Frequent if recurrence	Acute graft dysfunction, RPGN	Positive anti-GBM antibody within 6 months of transplant (transplant should be delayed) Alport syndrome may precipitate de novo anti-GBM disease post transplant	No established therapy Plasmapheresis, Cyclophosphamide, pulse steroids
IgA nephropathy	13-53%, reflecting varying threshold for biopsy	~10% over 10 years	Microscopic or gross hematuria, proteinuria, slow decline in graft function, RPGN. Typically occurs late post transplant.	Younger age Prior graft loss to recurrence Steroid-free maintenance regimen (controversial) No induction therapy Crescentic/rapidly progressive disease in native kidneys Living related donors (controversial)	No established therapy Consider ACEi or ARB for proteinuria, CNI, steroids Cyclophosphamide or rituximab if crescentic disease recurrence/RPGN Budesonide has not been studied in transplant recipients.
Lupus nephritis	2-9%, histologic recurrence in up to 30%	3% at 5-10 years	Typically recurs in first 10 years post transplant. Microscopic or gross hematuria, proteinuria, slow decline in graft function	Transplant while disease not quiescent; usually 6-12 months of disease quiescence Antiphospholipid syndrome is a risk of thrombosis	No established therapy MMF Anticoagulation if antiphospholipid syndrome before transplant
ANCA vasculitis	0-20%	6-8% at 5-10 years	Microscopic or gross hematuria, proteinuria, graft dysfunction	Transplant while disease not quiescent Positive ANCA serology does not appear to influence relapse.	No established therapy Consider cyclophosphamide, rituximab ± IVIg
MPGN	Immune complex: 20-33% DDD/C3GN: 67-100%	Immune complex: High DDD/C3GN: 34-66%	Proteinuria, hematuria, graft dysfunction. DDD occurs later than C3GN and usually isolated graft dysfunction. Low C3 levels	C3 glomerulopathy subtype Monoclonal gammopathy, lower serum complement levels, higher protein, crescents in native biopsy Rapid progression to ESKD	No established therapy Eculizumab for C3 glomerulopathy-limited success, cyclophosphamide, rituximab Treat underlying monoclonal gammopathy/infection

Continued on following page

Table_69_5_part1

Table 69.5 Recurrent Glomerulonephritis After Kidney Transplantation[274,277,416,417] (Continued)

Primary Disease	Recurrence Rate	Graft Loss With Recurrence	Presentation	Clinical Predictors of Recurrence	Treatment*
Membranous nephropathy	10-30%	10-50% at 10 years	Minimal proteinuria to nephrotic syndrome	Higher APLA2R titers Persistent elevated APLA2R titers	No established therapy Treatment of secondary factors Consider RAAS inhibition, rituximab, CNI, cyclophosphamide

[a]Treatment for most recurrent glomerulonephritis after kidney transplant is derived from studies in native kidney disease and utilized with limited data in transplant populations.
ACE, Angiotensin-converting enzyme; *ANCA,* antineutrophil cytoplasmic antibody; *Anti-GBM,* antiglomerular basement membrane; *APLA2R,* anti-phospholipase 2 receptor antibody; *ARB,* angiotensin receptor blocker; *C3GN,* C3 glomerulonephritis; *CNI,* calcineurin inhibitors; *DDD,* dense deposit disease; *ESKD,* end-stage kidney disease; *FSGS,* focal segmental glomerulosclerosis; *IVIG,* intravenous immunoglobulin; *MMF,* mycophenolate mofetil; *MPGN,* membranoproliferative glomerulonephritis; *RAAS,* renin-angiotensin-aldosterone system; *RPGN,* rapidly progressive glomerulonephriti.

Table_69_5_part2

Table 69.6 Routine Surveillance Laboratory Testing After Transplantation

	<1 Month	1-2 Months	2-6 Months	6-24 Months	>24 Months
Complete blood cell count	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Basic metabolic panel/glucose/phosphorus	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Drug level[a]	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Liver enzymes	Weekly	Weekly	Every 2 weeks	Monthly	Every 6-12 months
Urinalysis	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Lipid profile			Annually	Annually	Annually
Urine protein creatinine ratio[b]			3 Monthly	6 Monthly	Annually
BK polymerase chain reaction	Monthly	Monthly	3 Monthly	3/6 Monthly	
Parathyroid hormone[c]			3 Monthly[c]	6 Monthly[c]	Annually[c]
Donor-specific antibodies[d]	[d]	[d]	[d]	[d]	[d]

[a]Depending on the immunosuppressive regimen, to include tacrolimus, cyclosporine, mycophenolic acid.
[b]Frequent UPCR monitoring for patients with end-stage renal disease from focal segmental glomerulosclerosis.
[c]Posttransplant parathyroid hormone monitoring frequency determined by individual parathyroid hormone and calcium levels.
[d]Monitoring frequency determined by patient risk profile.
Standard target levels include:
- CsA (C0) 150-300 ng/mL early and 100-200 ng/mL late.
- CsA (C2) 1400-1800 ng/mL early and 800-1200 ng/mL late.
- Tacrolimus (C0) 8-12 ng/dL early and 5-8 ng/dL late.

Table_69_6

Table 69.7 Donor, Recipient, Donor-Recipient Paring, and Surgical Factors Associated With Outcomes After Kidney Transplantation

Donor Factors	
Living donor	Living donor allografts are superior to deceased donor allografts. Likely reflects: Very healthy living donors, the absence of brain death, benefits of elective vs. semiurgent surgery, minimization of ischemia-reperfusion injury, higher nephron mass.
DCD	Short-term outcomes (e.g., DGF, primary nonfunction) with DCD donors are inferior to those with NDD donors. Long-term outcomes of DCD organs (from donors <50 years old) are similar to those from standard deceased donors.
Age	Reduced allograft survival with extremes of donor age. Kidneys from older donors have fewer functioning nephrons due to aging and conditions such as hypertension and atherosclerosis. Younger donors (<5 years) are associated with worse outcomes, reflecting higher rates of technical complications.
Sex	Kidneys from female donors have slightly poorer survival probably due to nephron underdosing due to smaller kidney mass in women than men.
Race/Ethnicity	Deceased donor kidneys from AA are associated with reduced allograft survival largely relating to the presence of the APOL1 high-risk variant (in ~15% of AA donors). AA kidneys from donors lacking the APOL1 high-risk variant are comparable to non-AA donated kidneys. Integration of APOL1 genotyping (instead of race) into clinical prediction models (e.g., KDPI) is warranted.
Recipient Factors	
Timing	Preemptive transplantation is associated with less acute rejection and allograft failure. Longer time on dialysis is independently associated with poorer patient and graft survival.
Age	Graft survival rates are poorer in those at the extremes of age (<18 years; >65 years). In the very young, technical causes of graft loss (e.g., vessel thrombosis) and acute rejection are more common; death with a functioning graft is rare. In elderly recipients, death with a functioning graft is much more common (~ >50% of graft failures); acute rejection is less common.
Sex	Female recipients experience more graft loss and higher excess mortality than male recipients, particularly when the donor is male. Females are at higher rejection risk and have higher immune sensitization than males (primarily due to pregnancy).
Race/Ethnicity	AA recipients have worse graft survival than white recipients reflecting multiple factors including more DGF, more acute and late rejection, stronger immune responsiveness, a predominantly white donor pool (with resultant poorer matching of HLA and non-HLA antigens), altered pharmacokinetics of immunosuppressive drugs, differences in socioeconomic factors, and a higher prevalence of hypertension. Asian and Latino recipients have superior outcomes to white recipients; it is unknown why.
PRA	Patients who are highly sensitized are at higher risk for adverse early and late graft outcomes compared with unsensitized recipients.
DGF	DGF is associated with poorer patient and allograft survival. Allograft half-life is reduced by 30% with DGF, a larger effect than early acute rejection.
Compliance	Poor compliance with the immunosuppressive regimen markedly increases the risk of acute rejection (particularly late acute rejection) and allograft loss. Allograft loss has been reported to be 7-fold higher in nonadherent patients.
Donor-Recipient Factors	
HLA MM	Greater degree of HLA mismatch increases rejection risk and reduces allograft survival. HLA epitope-matching may prove important in coming years, though not widely used at present.
CMV matching	CMV D–/R– pairings have the best outcomes, D+/R– pairings have the worst. CMV probably affects graft outcomes through overt infection, but subclinical effects on immune function may also be important.
Sex matching	Female recipients of male kidneys have worse graft survival relating to an immune response to antigens on the Y-chromosome (H-Y antigens). The generally larger nephron dose in male donors may be a confounding factor in registry analyses.
Size matching	An imbalance between the metabolic demands of the recipient and the functional transplant mass (nephron load) may contribute to development and progression of chronic allograft injury when the recipient size is larger than the donor (>30 kg)
Surgical Factors	
Center effect	Outcomes vary from transplant center to center reflecting normal statistical variance, as well as center expertise and culture.
CIT	Prolonged CIT is associated with higher risk of DGF and poorer allograft survival.
WIT	Prolonged WIT is associated with higher risk of DGF and poorer allograft survival.

AA, African Americans; *CIT,* cold ischemic time; *CMV,* cytomegalovirus; *D,* donor; *DCD,* donation after cardiac death; *DGF,* delayed graft function; *HLA,* human leukocyte antigen; *KDPI,* kidney donor profile index; *NDD,* neurologic determination of death; *PRA,* panel reactive antibody; *R,* recipient; *WIT,* warm ischemic time.

Table_69_7

Table 69.8 Cardiovascular Risk Factor Management in Kidney Transplant Recipients[175,300,305,416-437]

Risk Factor	Goal of Treatment	Treatment Recommendation	Comments
Posttransplant diabetes	Target HbA1c 7.0-7.5%; avoid HbA1c ≤6.0% especially if hypoglycemia reactions are common	Lifestyle modification; oral agents; insulin	Avoid metformin with reduced eGFR due to risk of lactic acidosis; Emerging data on SGLT-2 inhibitor use in transplant recipients are pending
Hypertension	<130/<80 mm Hg	Lifestyle modification; calcium channel blockers are first line; ACEi/ARB if proteinuria, diabetes of CVD; β-blocker if CVD	Nondihydropyridine CCB (verapamil and diltiazem) interact with CNIs; elevated CNI levels often require dose adjustment
Proteinuria	Reduce proteinuria	Lifestyle modification; when urine protein excretion ≥1 g/day for ≥18 years old and ≥600 mg/m^2/24 h consider ACEi/ARB as first line	Clinically important reductions in proteinuria with RAAS blockade, but no clear evidence of reduced all-cause mortality, transplant failure, or doubling of serum creatinine
Dyslipidemia	Adults: total cholesterol <200; LDL <100; non-HDL <130; Triglycerides <150 Adolescents: LDL<130; Non-HDL <160	Lifestyle modification; statins; fibrates; ezetimibe Patients with a kidney transplant should receive statin therapy, irrespective of lipid profile	Starting dose of a statin should be reduced as CNIs (especially cyclosporine) increase statin blood—increased risk of statin-related toxicities
Obesity	BMI <35-40 kg/m^2	Lifestyle counseling including an assessment of psychosocial stressors, sleep hygiene Increased physical activity and caloric restriction; bariatric surgery	Increased complications with bariatric surgery post transplant indicate more research needed; emerging practice to use GLP-1 RA (data on risk/benefit in transplant recipients are pending)
Tobacco use	Smoking cessation	Psychologic and/or pharmacologic support	Risk of weight gain, lifestyle counselling as above.

ACEi, Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker; *BMI,* body mass index; *CVD,* cardiovascular disease; *GLP-1 RA,* glucagon-like peptide-1 receptor agonists; *HbA1c,* hemoglobin A1c; *HDL,* high-density lipoprotein; *LDL,* low-density lipoprotein; *SGLT-2,* sodium-glucose cotransporter 2.
From Kidney Disease: Improving Global Outcomes (KDIGO) Transplant Work Group. KDIGO clinical practice guideline for the care of kidney transplant recipients. *Am J Transplant*. 2009;9(Suppl 3):S1–S155.

Table_69_8

Table 69.9 Antihypertensive Agents in the Transplant Recipient

Indication	β-blocker	ACE-I/ARB	Ca-Channel Blocker	Diuretic Thiazide
Hypertension only		X	X	X
CHF	X	X		
Post-MI	X	X		
CAD	X	X		
Diabetes		X		
Proteinuria		X		

ACE-I, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; CAD, coronary artery disease; CHF, congestive heart failure; MI, myocardial infarction.

Table_69_9

Table 69.10 Cancers Categorized by Standard Incidence Rate (SIR) for Kidney Transplant Patients and Cancer Incidence

	Common Cancers in Both General and Transplant Populations: Incidence >10 Per 100,000 People	Common Cancers in Transplant Population: Incidence in General Population <10 Per 100,000, but Estimated Incidence in Transplant Population >10 Per 100,000 People	Rare Cancers Incidence in Both General and Transplant Populations <10 Per 100,000 People
High SIR >5	Kaposi sarcoma (with HIV)	Kaposi sarcoma Vagina Non-Hodgkin lymphoma Kidney Nonmelanoma skin Lip Thyroid Penis Small intestine	Eye
Moderate SIR (1-5)	Lung Colon Cervix Stomach Liver	Oronasopharynx Esophagus Bladder Leukemia	Melanoma Larynx Multiple myeloma Anus Hodgkin lymphoma
No Increased Risk	Breast Prostate Rectum		Ovary Uterus Pancreas Brain Testis

Adapted from the KDIGO practice gudelines.

Table_69_10

Table 69.11 Viral Infections Associated With Development of Cancers in Kidney Transplant Patients

Virus	Neoplasm
HBV	Hepatocellular cancer
HCV	Hepatocellular cancer
EBV	PTLD
HPV	Squamous cell cancers of anogenital area and of mouth
HHV-8	Kaposi sarcoma

EBV, Ebstein-Barr virus; *HBV,* hepatitis B virus; *HCV,* hepatitis C virus; *HHV-8,* human herpes virus-8; *HPV,* human papillomavirus; *PTLD,* posttransplant lymphoproliferative disorder.

Table_69_11

Table 69.12 Clinical and Pathologic Spectrum of Posttransplant Lymphoproliferative Disorder (PTLD) and Summary of its Management

	Early Disease (50%)	Polymorphic PTLD (30%)	Monoclonal PTLD (20%)
Clinical features/ symptoms	Infectious mononucleosis-type illness	Infectious mononucleosis-type illness ± weight loss, localizing symptoms	Fever, weight loss, localizing symptoms
Pathology	Preserved architecture, atypical cells infrequent	Intermediate transformed cells	High-grade lymphoma with confluent and marked atypia
Clonality	Polyclonal	Usually polyclonal	Monoclonal
Treatment	Reduce immunosuppression	Reduce immunosuppression ±rituximab	Reduce immunosuppression ± rituximab ± chemotherapy
Prognosis	Good	Intermediate	Intermediate/poor

PTLD, Posttransplant lymphoproliferative disorder.

Table_69_12

Table 69.13 Manifestations of Cytomegalovirus Disease in the Kidney Transplant Recipient

Tissue Affected	Clinical/Pathologic Features	Comment
Viremia (blood)	Fever, malaise, myalgia	Nonspecific, may occur after cessation of prophylaxis
Bone marrow	Leukopenia	May require azathioprine or MMF dose reduction. Valganciclovir may also cause leukopenia
Lungs	Pneumonitis	May be life-threatening; exclude coinfection with other organisms
GI tract	Inflammation and ulceration of esophagus or colon	May be life-threatening
Liver	Hepatitis	Rarely severe
Retina	Blurred vision, flashes, floaters	Relatively rare in kidney transplants; if occurs, usually late.
Kidney	Viral inclusions	Association with FSGS

GI, Gastrointestinal; MMF, mycophenolate mofetil.

Table_69_13

Table 69.14 Recommended and Contraindicated Vaccines in Transplant Recipients

Recommended	Comments	Contraindicated[a]
Influenza	Annually	Live varicella vaccine[b]
PCV13	Once	BCG
PPSV23	Repeat every 5 years, up to 3 doses/lifetime including one dose in those aged 65 years and older	Smallpox
Tdap/Td	Td booster every 10 years	Intranasal influenza
9vHPV	Only for ≤ 26 years of age	Live Japanese B encephalitis
HepA		Oral polio
HepB	Give 3-dose series if anti-HBS<10 mIUL/mL	MMR
Shingrix	Adults 50+ years old; 2 doses 2-6 months apart	
COVID-19	As per evolving local guidelines	
Asplenia or complement inhibitor therapy		Live oral typhoid Ty21a
MenACWY	2-dose series with booster every 5 years	Yellow fever
MenB	2-dose series, no booster	
Hib	If not already given	

[a]Not an exhaustive list, all live vaccines are generally contraindicated.
[b]Awaiting data on the non-live "Shingrix" (HZ/su) zoster vaccine.
9vHPV, Human papillomavirus vaccine; *COVID-19, Sars-CoV-2 vaccine (undefined); HepA,* hepatitis A vaccine; *HepB,* hepatitis B vaccine; *Hib, Haemophilus influenzae* type b conjugate vaccine; *MenACWY,* meningococcal (quadrivalent) conjugate; *MenB,* serogroup B meningococcal vaccines; *MMR,* measles, mumps, and rubella vaccine; *PCV13,* Pneumococcal conjugate vaccine (13-valent); *PPSV23,* pneumococcal polysaccharide vaccine (23-valent); *Shingrix,* herpes zoster vaccine; *Td,* tetanus and reduced diphtheria toxoids; *Tdap,* tetanus and reduced diphtheria toxoid, and acellular pertussis vaccine.

Table_69_14

eTable 69.1 Common Cold Storage Preservation Solutions

Solution	Minimization of Intracellular Edema
UW	Viscosity 3x water; composition similar to intracellura fluid (potassium meq/L); lactobionate and raffionose prevent cellular edema and hydroxyethyl first buffers free radicals.
HTK	Low viscosity, low sodium and potassium (Both 15 mmol/L); Histidine serves buffer; Tryptophan and mannitol are free radical scavengers; low cost

UW, University of Wisconsin; *HTK,* Histidine-Tryptophan-Ketoglutarate.

Table_e69_1

eTable 69.2 Toxicity Profiles of Immunosuppressive Medications

		Calcineurin Inhibitors		Antiproliferative Agents		
Adverse Effect	Steroids	CsA	Tac	MMF	AZA	mTORi
Posttransplant diabetes	↑	↑	↑↑			↑
Dyslipidemia	↑	↑				↑↑
Hypertension	↑↑	↑↑	↑			
Osteopenia	↑↑	↑	(↑)			
Anemia and leukopenia				↑	↑	↑
Delayed wound healing						↑
Diarrhea, nausea/vomiting			↑	↑↑		↑
Proteinuria						↑↑
Decreased eGFR		↑↑	↑			
Hair changes		↑	↓			
Gingival hyperplasia		↑				
Peripheral edema	↑					↑

AZA, Azathioprine; *CsA,* cyclosporine A; *MMF,* mycophenolate mofetil; *mTORi,* mammalian target of rapamycin inhibitor(s); *Tac,* tacrolimus.
↑ Indicates a mild-moderate adverse effect of the complication.
↑↑ Indicates a moderate-severe adverse effect of the complication.
(↑) Indicates a possible, but less certain adverse effect on the complication.
Adapted from KDIGO clinical practice guideline for the care of kidney transplant recipients. *Am J Transplant.* 2009;9(Suppl 3):S1–155.

Table_e69_2

eTable 69.3 Noninvasive Diagnostic Testing for Acute Allograft Rejection in Kidney Transplant Recipients

- Donor-specific antibodies (DSA)
- Donor-derived cell-free DNA (dd-cfDNA)
- Blood gene expression profiles (Trugraf)
- Urinary mRNA
- Urinary levels of chemokine
- Proteomic and peptidomic signatures of acute rejection in urine and blood samples
- IFN-γ enzyme-linked immunosorbent spot (ELISPOT) assay

From Thongprayoon C, Vaitla P, Craici IM, et al. The use of donor-derived cell-free DNA for assessment of allograft rejection and injury status. *J Clin Med.* 2020;9(5):1480. Published 2020 May 14.

Table_e69_3

eTable 69.4 Precautions for Procedures and Surgery in Kidney Transplant Recipients

- Caution with radiocontrast exposure
- Maintain hydration
- Avoid nephrotoxic antibiotics and analgesic
- "Stress-dose steroids" is not always necessary
- If enteral route of medication is contraindicated, give tacrolimus sublingually if able (1/2 total oral dose)
- If enteral route of medication is contraindicated/sublingual tacrolimus is not available, give calcineurin inhibitor via intravenous route (1/3 total oral dose for cyclosporine; 1/5 total oral dose for tacrolimus)
- Monitor allograft function, plasma potassium, and acid-base balance daily
- Consider wound-healing impairment with sirolimus

Table_e69_4